	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2025 – 2026*

Presentation on Capstone Project Execution (CPE–316004)

Topic	Text Document Categorization using ML	
Presented by	1. Monishka Nitin Sonawane	
	2. Mayuri Anil Torawane	
	3. Dia Dinesh Waswani	
	4. Yogeshwari Sudhir Chaudhari	
Guided by	Mr. C.P. Bhamare	
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Abstract

The Text Document Categorization System is a web-based application that automatically classifies text documents into different categories such as Sports, Technology, Education, Healthcare and others. Users can upload a ZIP file containing multiple documents, and the system processes each file to determine its most relevant category. The classification is performed using TF-IDF for text feature extraction and the Multinomial Naive Bayes algorithm for prediction. The system is developed using the Django framework with Python and integrates machine learning to automate document organization. This solution reduces manual effort and improves efficiency.

Content

1. Introduction (background of the industry or used based problem / task)
2. Literature Survey
3. Scope of the project
4. Methodology
5. Details of designs, working and processes
6. Results and applications
7. Conclusion and future scope
8. References and Bibliography

1. Introduction

In today's digital world, a huge amount of unstructured data such as documents, emails, and articles is generated every day. Text categorization is a technique in data mining that automatically assigns predefined categories to text documents using machine learning algorithms. It helps in efficiently organizing data and improving information retrieval. Feature selection plays a crucial role in text classification by reducing unnecessary data and improving processing speed. Traditional methods based only on word count may result in lower accuracy. Machine learning algorithms like Naïve Bayes can improve classification performance and accuracy.

1. Introduction

The effectiveness of the system also depends on the amount and quality of data used for training. Managing and organizing this data manually is difficult, time-consuming, and error-prone.

Machine Learning : Machine Learning is a branch of Artificial Intelligence that allows systems to learn from data and make predictions without being explicitly programmed.

1.Supervised Learning.

2.Unsupervised Learning.

3. Reinforcement Learning

4.Semi- Supervised Learning

1. Introduction

- **20 Newsgroups Dataset :** The system is trained using the 20 Newsgroups dataset, which contains thousands of categorized news articles across multiple topics.

- **Algorithms Used :**

TF-IDF = TF-IDF(t,d)=TF(t,d)×IDF(t)

Naive Bayes = P(C|X)=P(X|C)·P(C)/ P(X)

- **Model Training :** The machine learning model is trained on labeled text data from the 20 Newsgroups dataset and saved using joblib for reuse.

2. Literature Review

1. Study of Existing System

- Manual Document Categorization
- Problem with Traditional Systems
- Use of NLP Techniques
- Common Algorithms Used
- Datasets for Training
- Limitations of Existing Systems : Many existing systems are complex and not easily accessible for general users.

2. Literature Review

2. Problem Statement

With the rapid growth of digital information, organizations and individuals deal with large amounts of textual documents such as news articles, reports, and research papers. Manually categorizing these documents is time-consuming, inefficient, and prone to human error.

2. Literature Review

3. Problem Definition

The proposed system is a web-based news classification application that automatically categorizes news articles into areas such as Politics, Sports, Business, Technology, and Entertainment. It eliminates manual sorting, which is slow due to the large volume of news data. The system uses machine learning algorithms like Naive Bayes, SVM, and Logistic Regression to analyze text and predict the correct category. It is trained on the 20 Newsgroups dataset to ensure fast and accurate classification through an easy-to-use web interface.

3. Scope of the project

- **Key Scope Points**

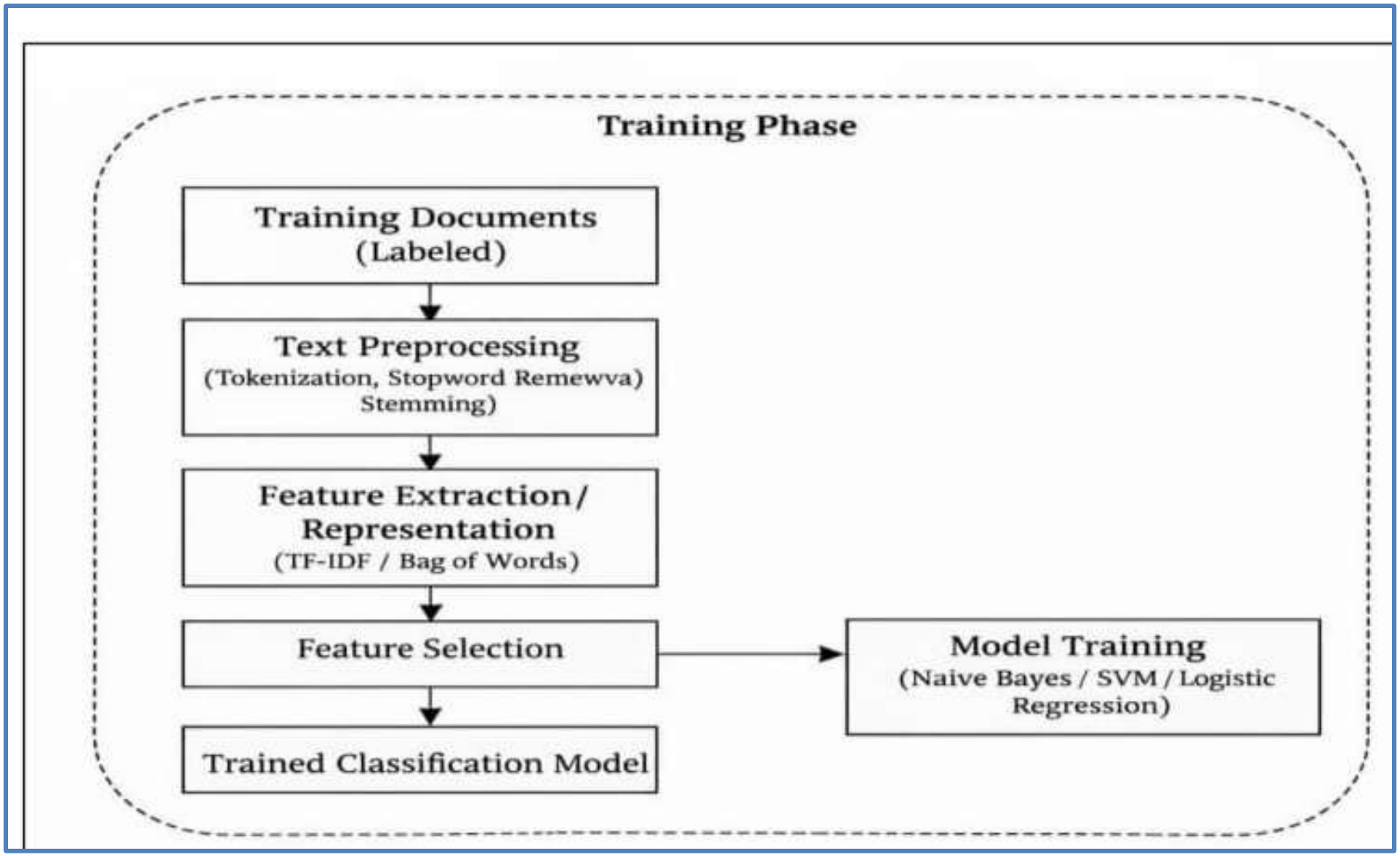
1. Automatic Document Classification
2. Machine Learning-Based Prediction
3. Dataset-Based Model Training
4. Web-Based Application
5. Document Management and Visualization

4. Methodology

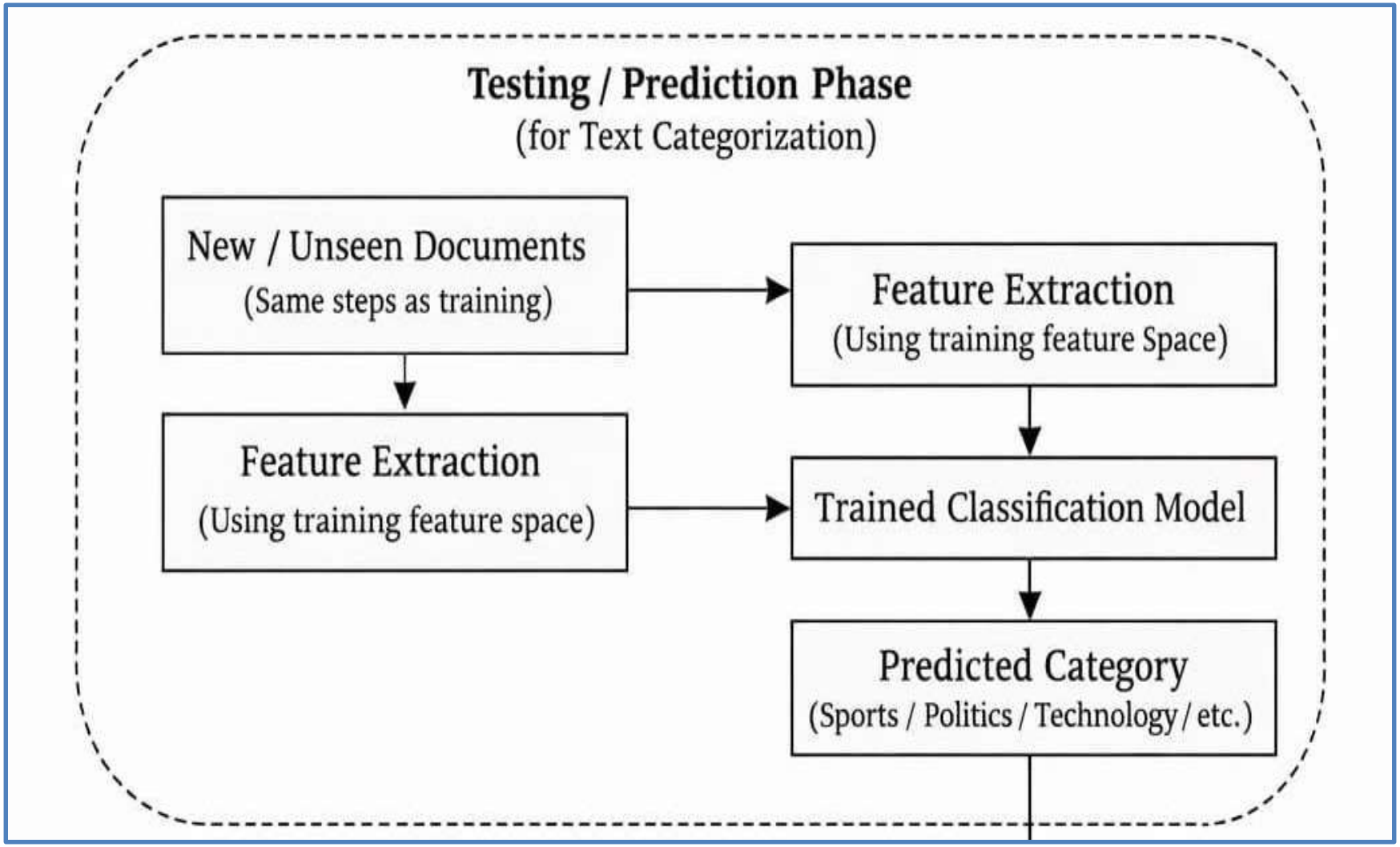
- **How the System Work**

1. Document Upload
2. Text Extraction
3. Text Preprocessing
4. Feature Extraction
5. Model Prediction
6. Result Display

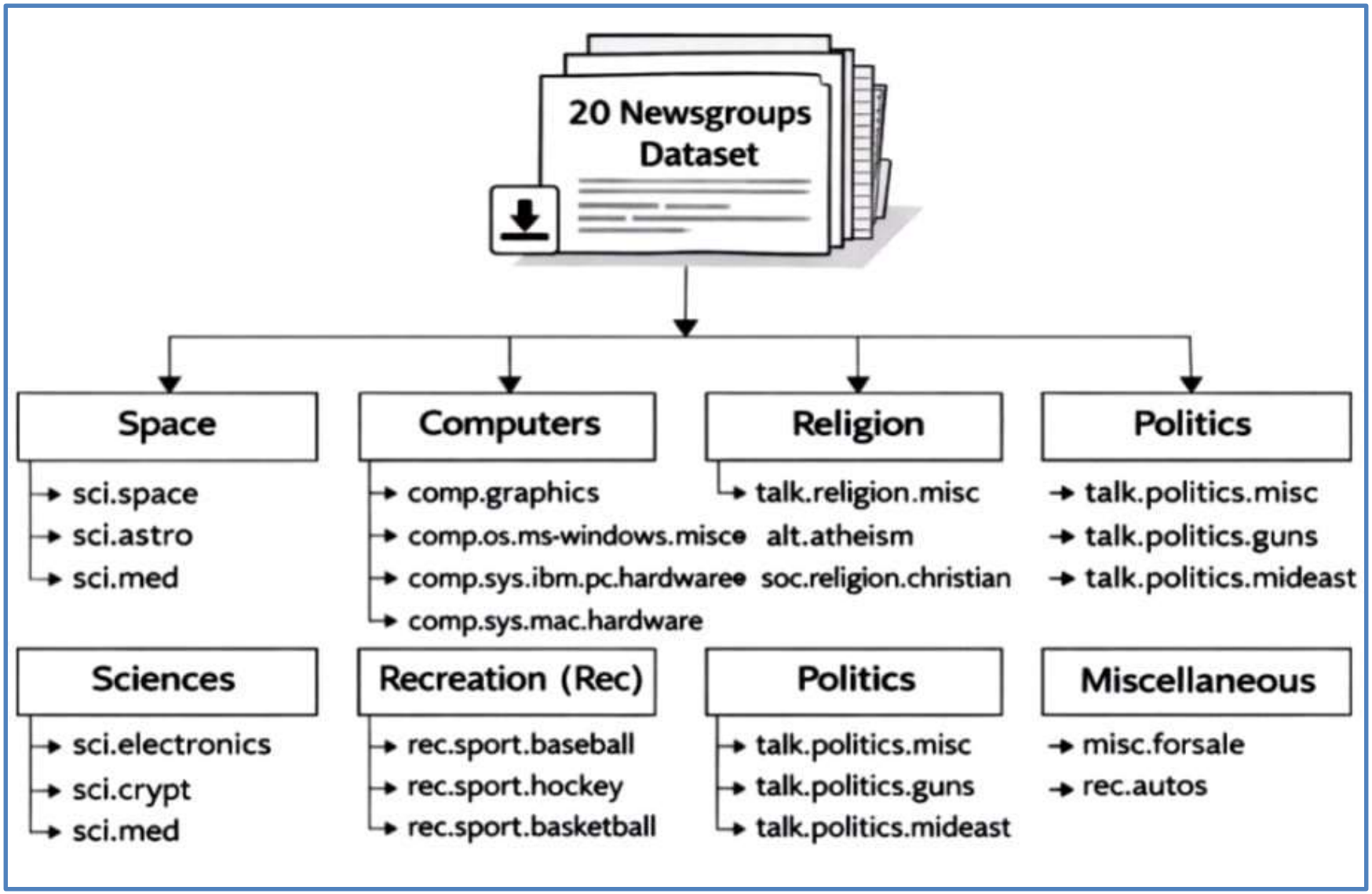
4. Methodology-Training Phase



4. Methodology-Testing Phase



4. Methodology-20 News Dataset



5. Details of designs, working & processes

System Design

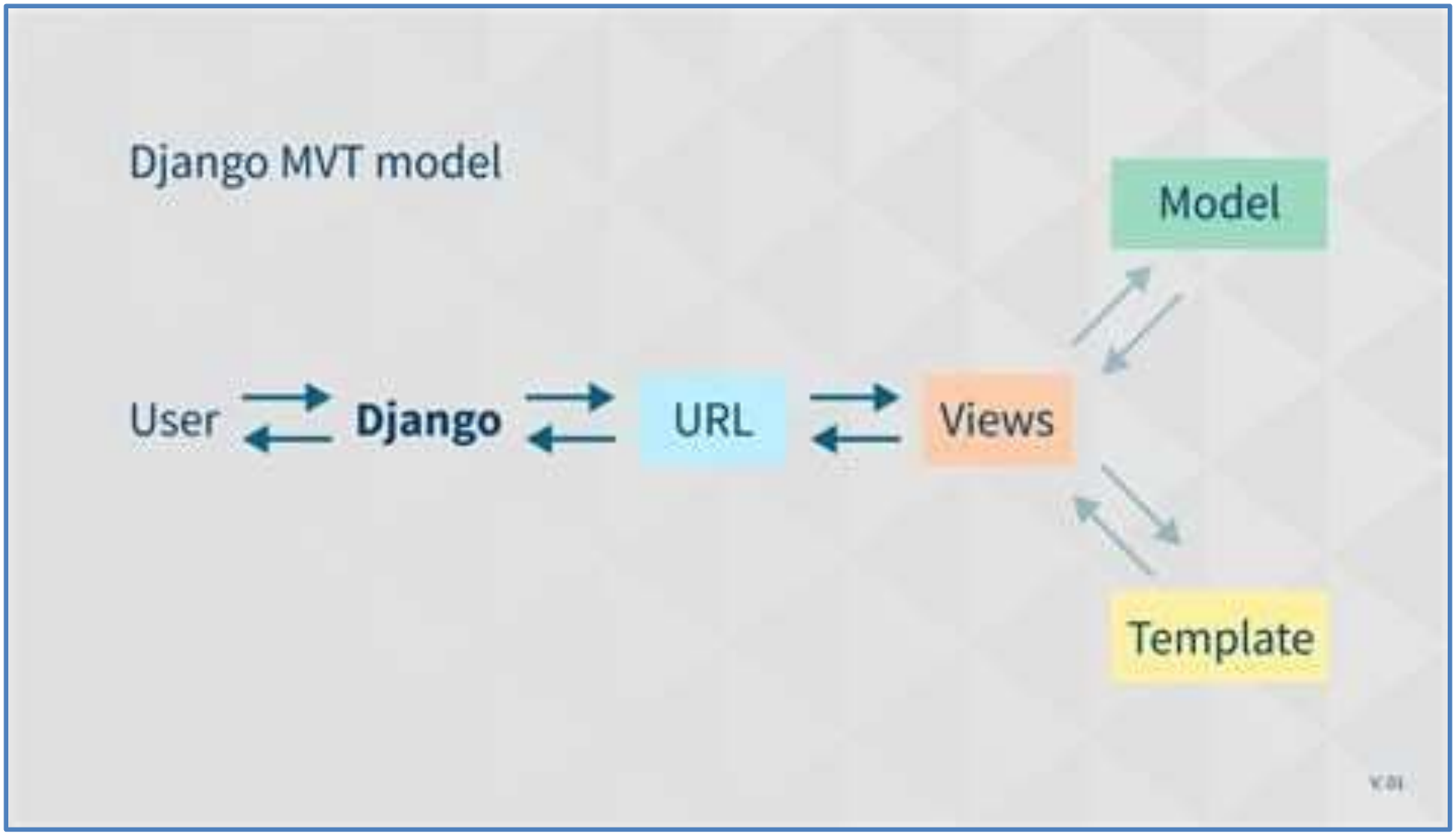
- The system is a web-based application built using Django.
- It follows MVT architecture (Model–View–Template).
- The Document model stores classification results in the database.
- Machine Learning logic is implemented in the views layer.
- Frontend templates display categorized results dynamically.

5. Details of designs, working & processes

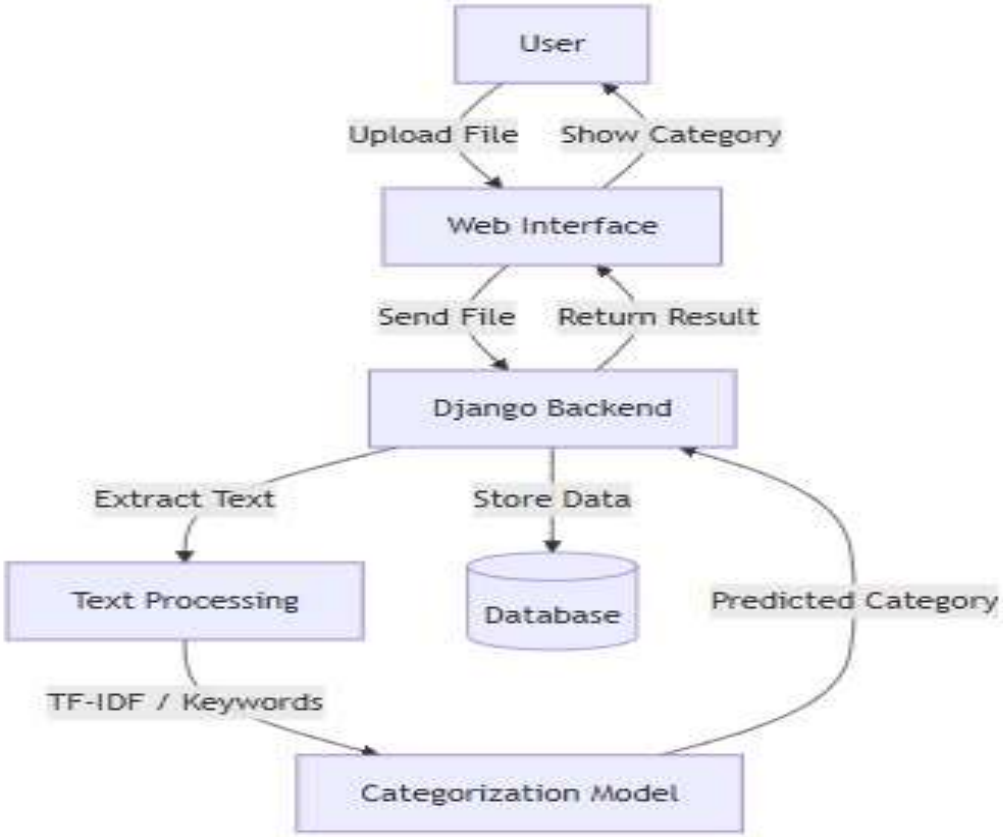
Working:

- The system accepts bulk documents in ZIP format.
- Each file is processed individually.
- Text analysis is performed using TF-IDF.
- Similarity comparison is done using Cosine Similarity.
- The predicted category and keywords are saved automatically.
- Users can view results category-wise through the dashboard.

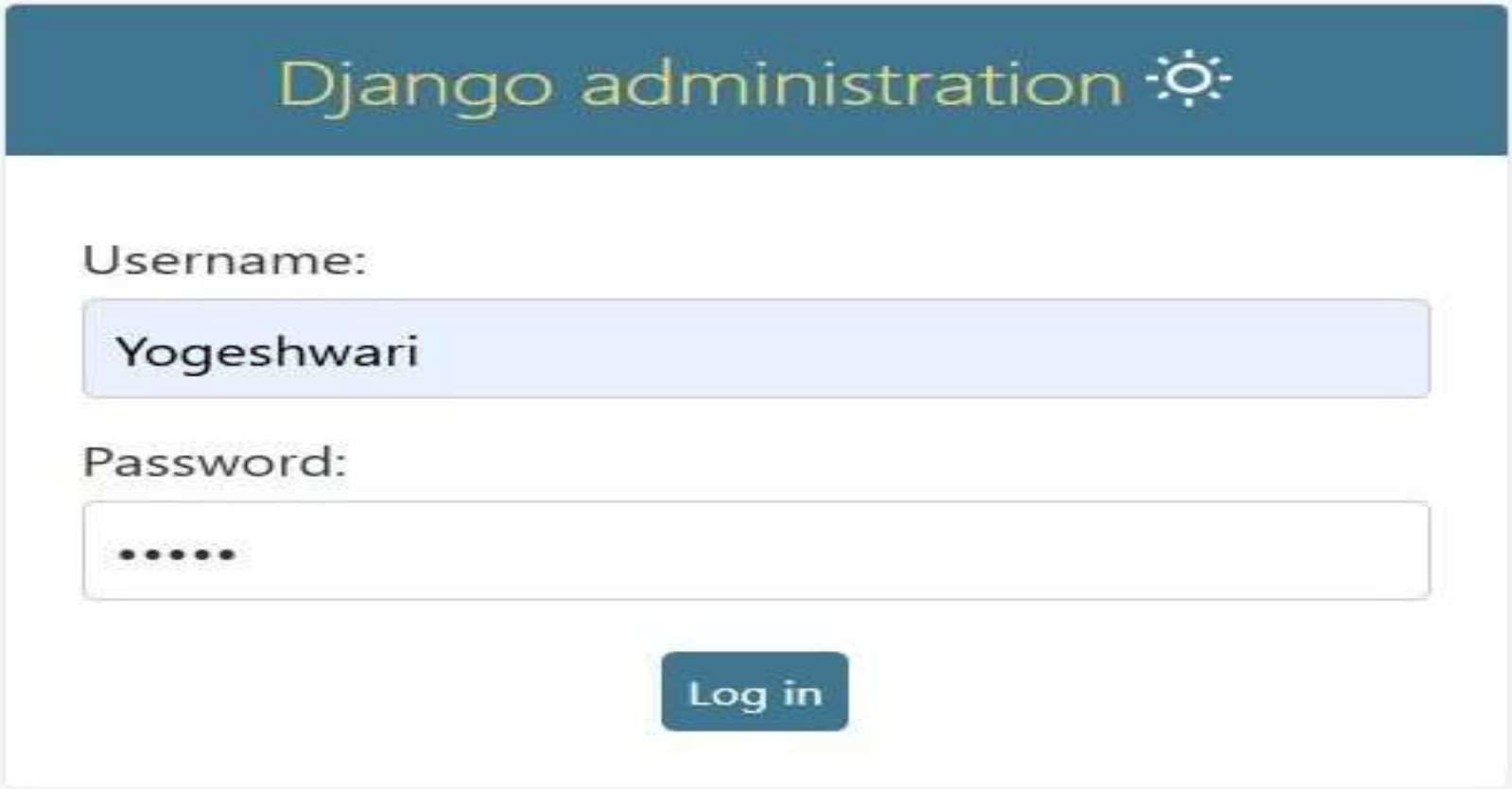
5. Details of designs, working & processes- MVT Architecture of Django



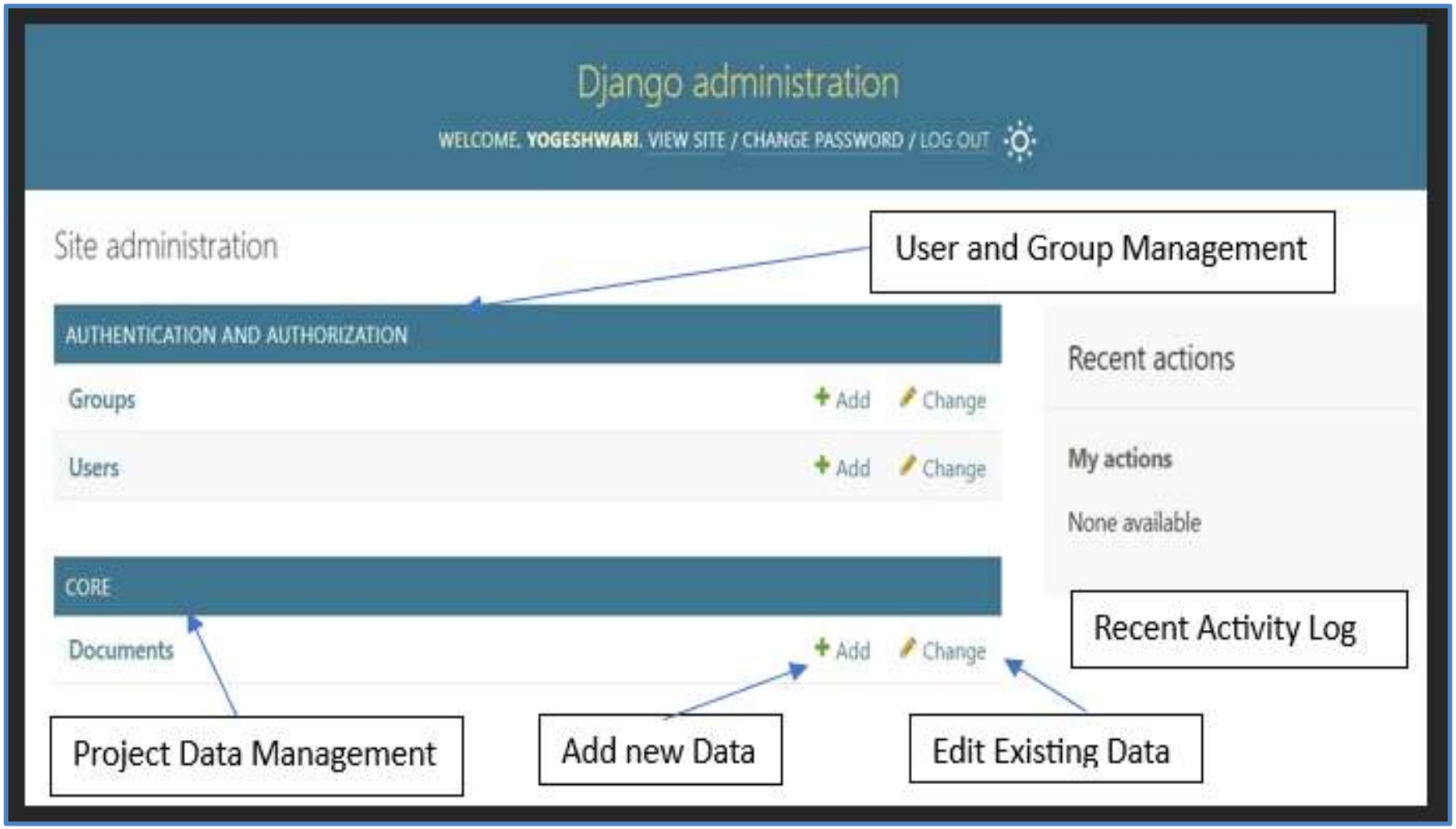
5. Details of designs, working & processes- Flow of the System



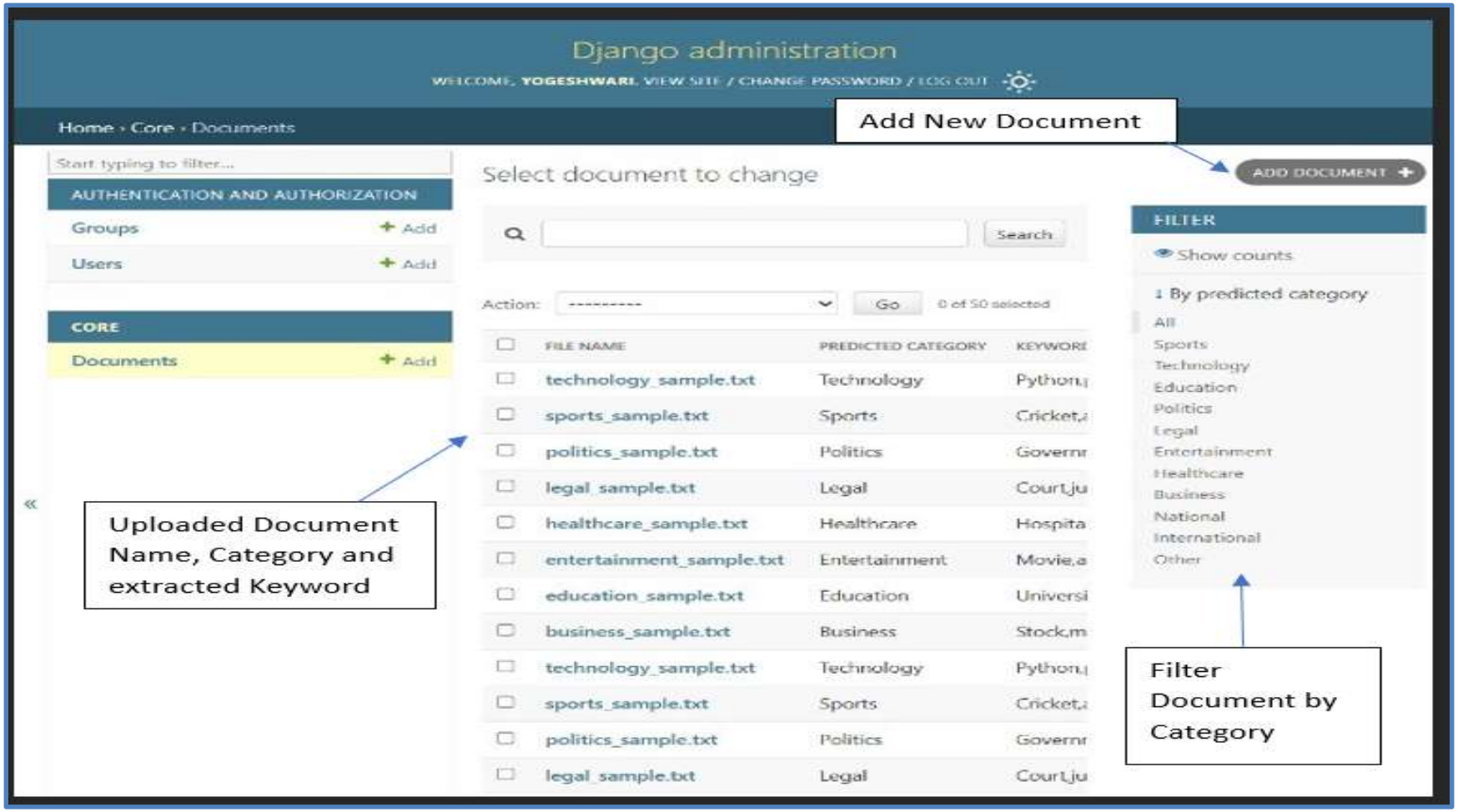
5. Details of designs, working & processes- Django Admin Login Panel



5. Details of designs, working & processes- Django Admin Dashboard



5. Details of designs, working & processes- Document Management Dashboard



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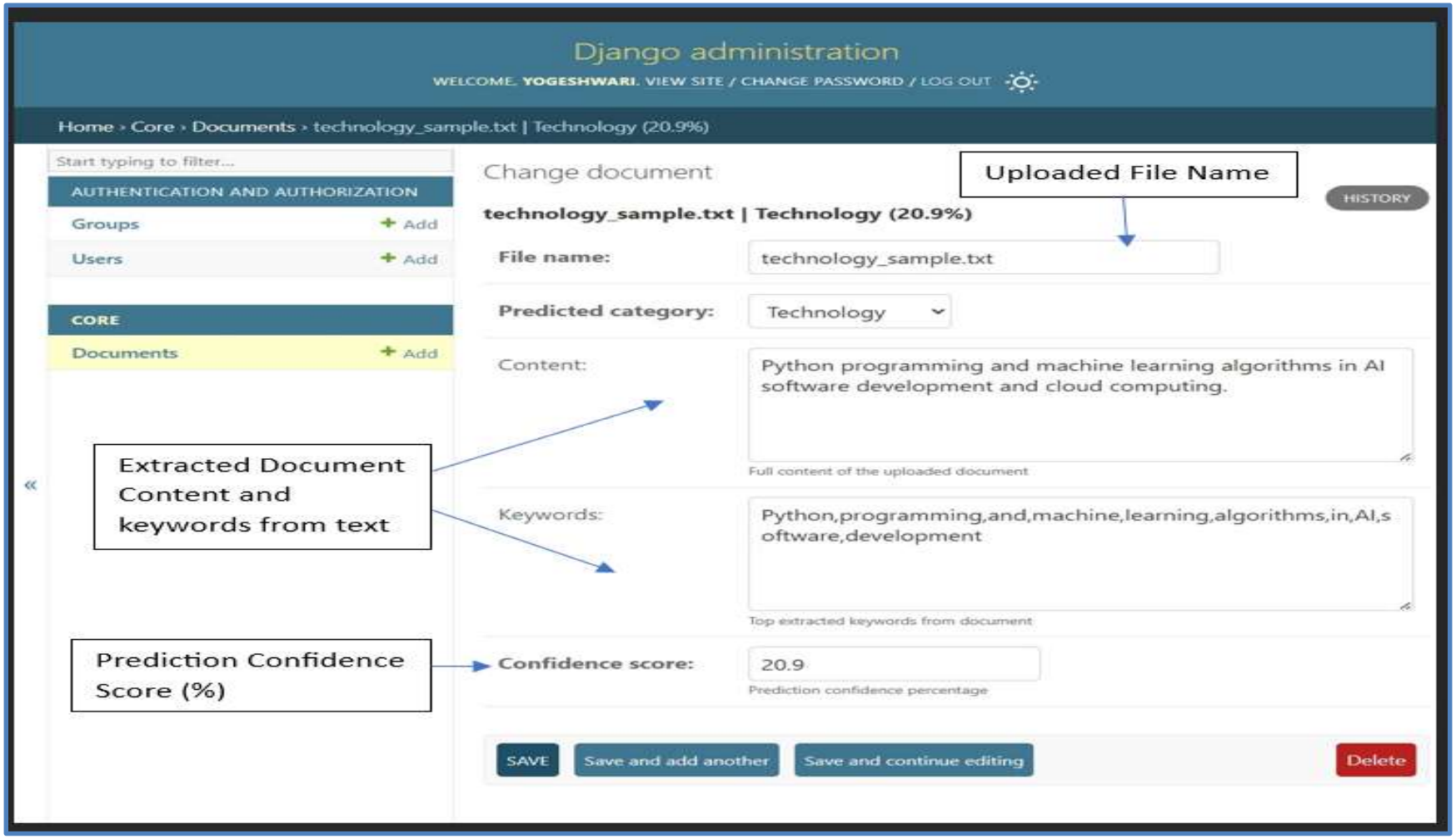

6. Results and Applications

Results:

- The system successfully categorized uploaded text documents into predefined categories.
- Multiple documents were processed automatically through ZIP file upload.
- Important keywords were extracted from each document.
- The dashboard displayed category-wise document counts clearly.
- The classification process was fast and efficient.

6. Results and Applications

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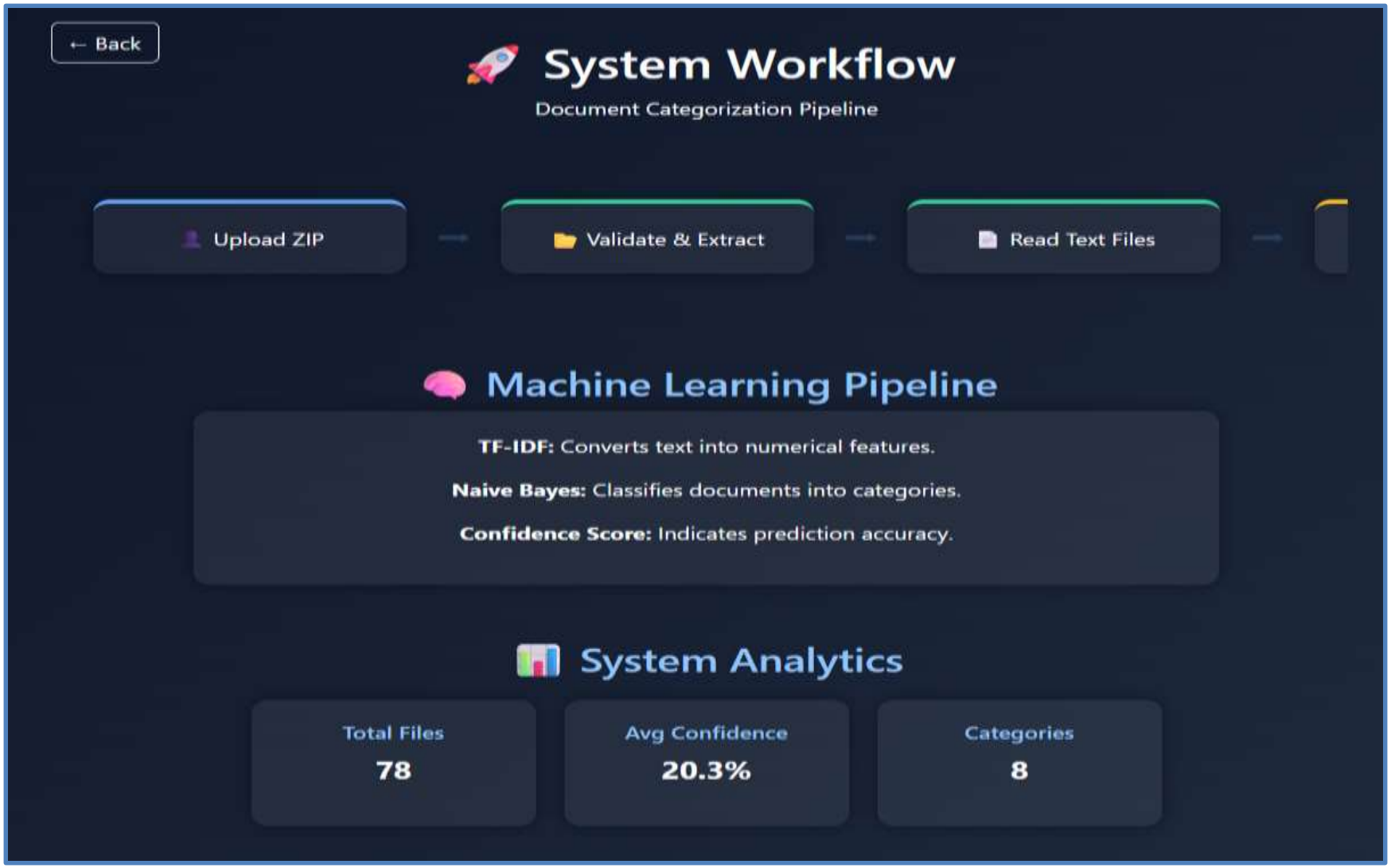

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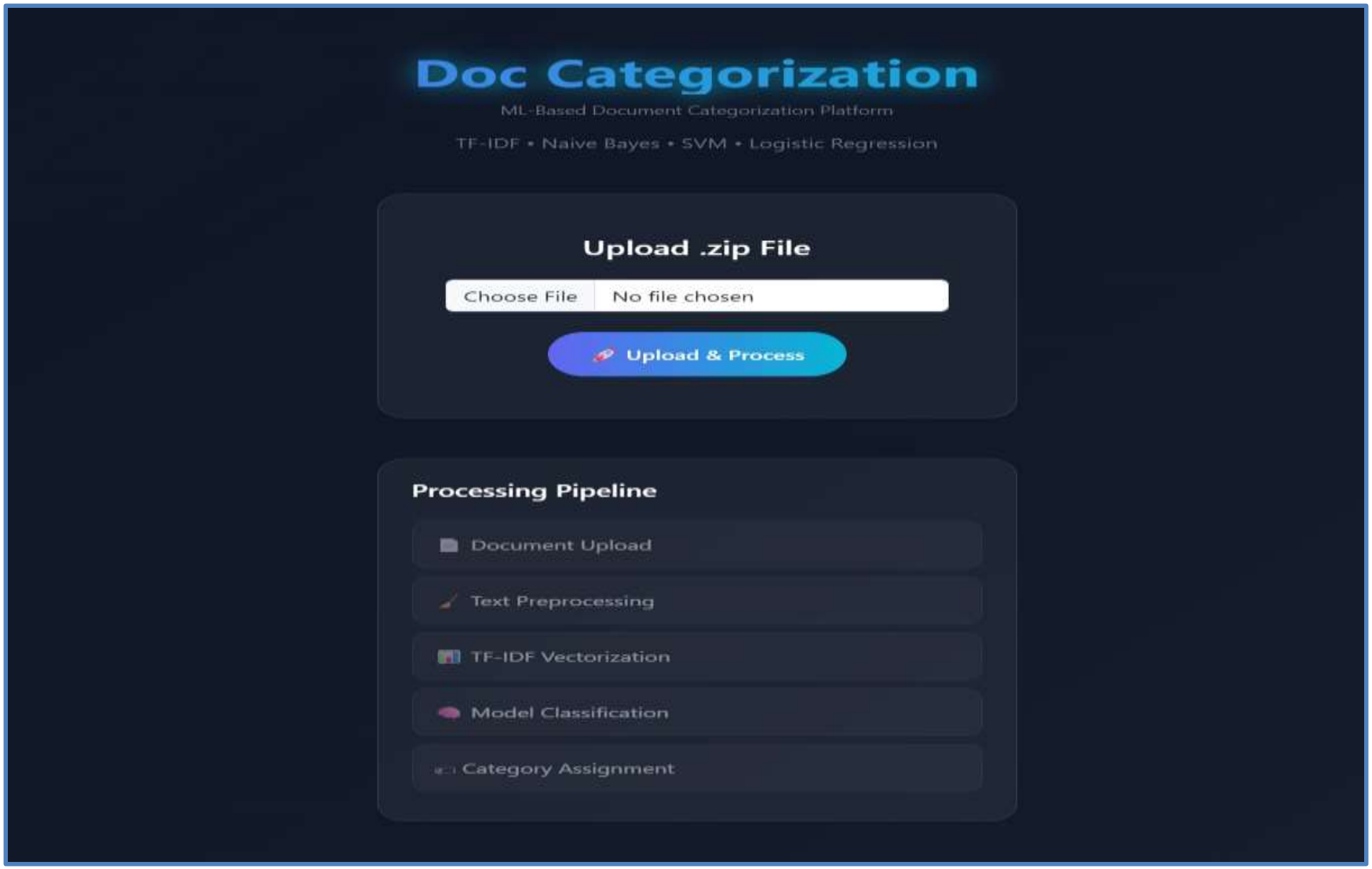

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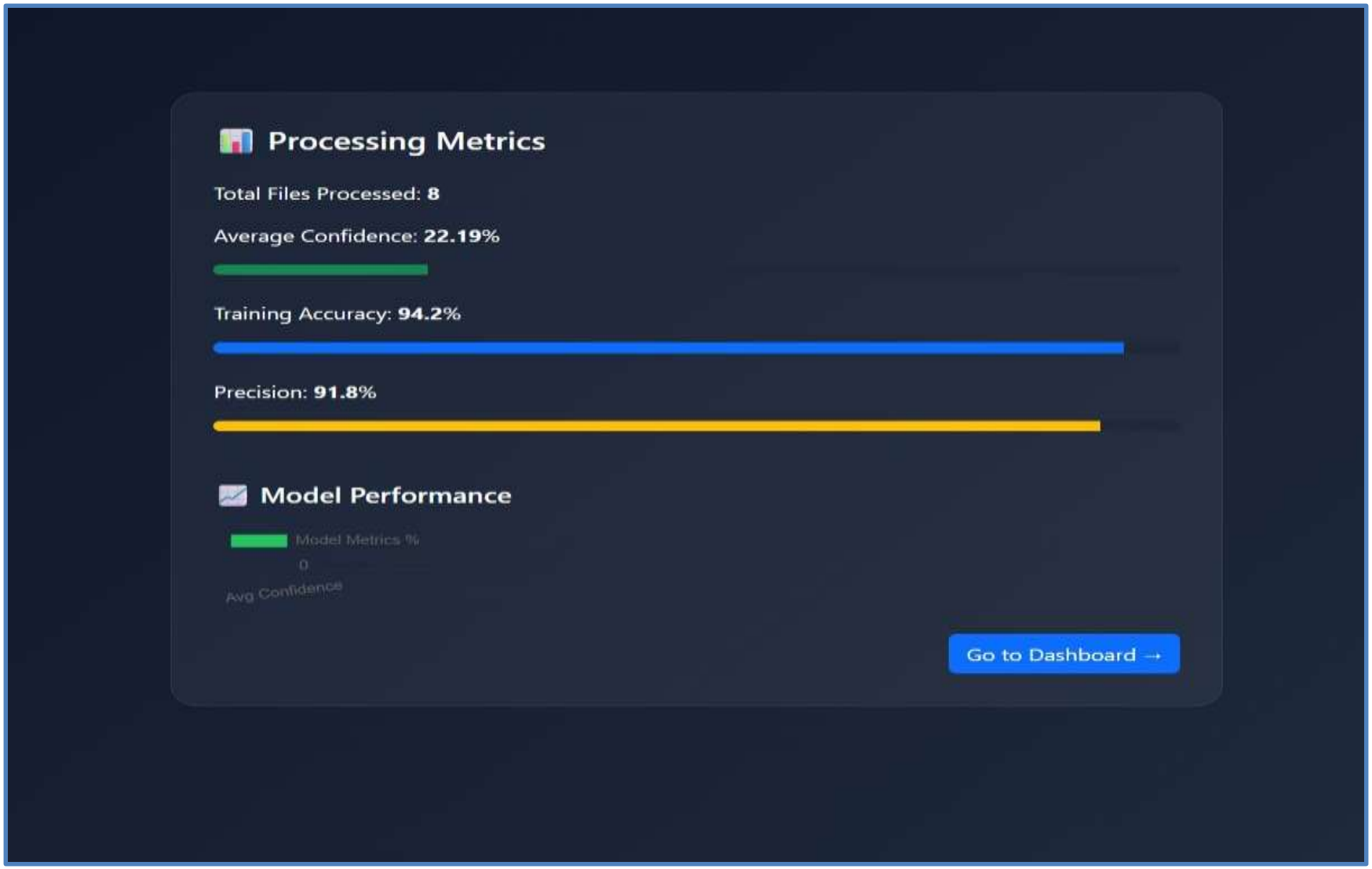

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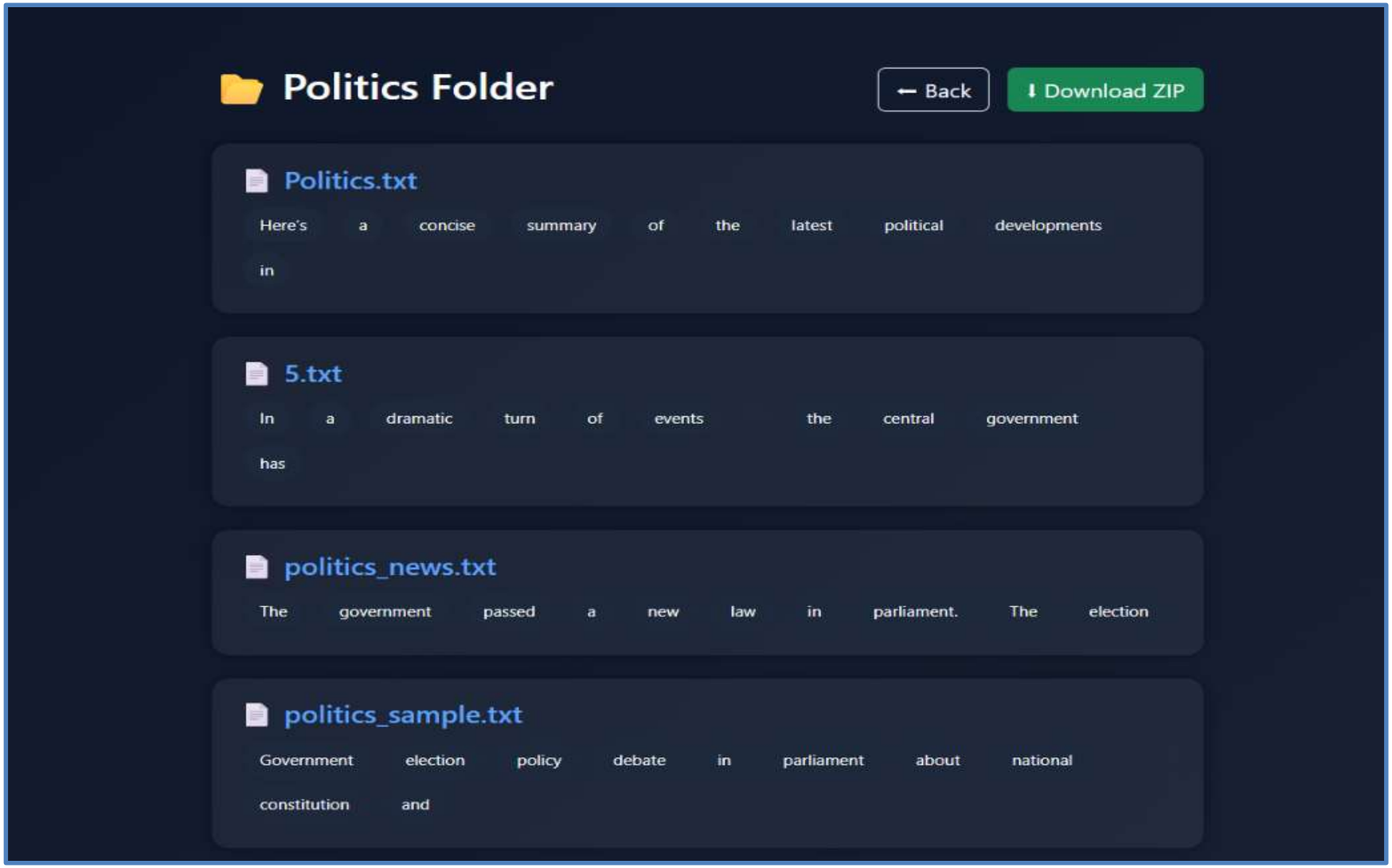

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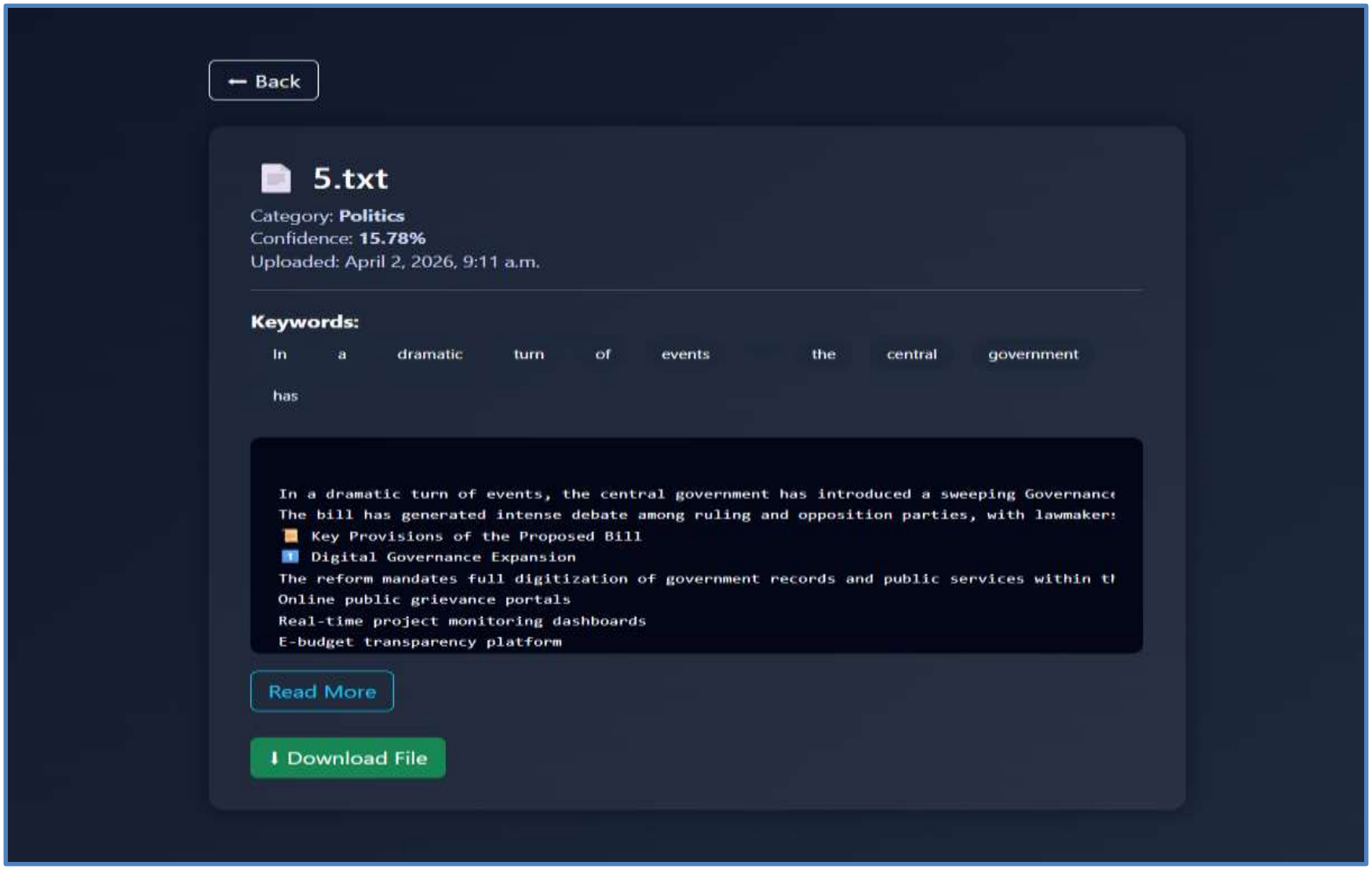

6. Results and Applications

Applications:

- News article classification
- Email filtering systems
- Research paper organization
- Legal document management
- Content management systems
- Educational document sorting

7. Conclusion and future scope

Conclusion:

- The project successfully implements an automatic Text Document Categorization system using Django and Python.
- It uses TF-IDF and Cosine similarity to classify documents into predefined categories.
- The system efficiently processes multiple files and reduces manual effort in organizing text data.

7. Conclusion and future scope

Future Scope:

- Implement advanced Deep Learning models like LSTM or BERT for higher accuracy.
- Add support for PDF and Word documents.
- Improve accuracy using a trained dataset.
- Develop a REST API for integration with other systems.
- Deploy the application on a cloud server for real-time use.

8. References and Bibliography

- Python**:** https://www.python.org/
- Django: https://docs.djangoproject.com/
- TF-IDF: https://en.wikipedia.org/wiki/Tf%E2%80%93idf
- Multinomial Naïve Bayes :

  https://en.wikipedia.org/wiki/Naive_Bayes_classifier